

Supplementary File S5

RQ5: Validation Maturity, Deployment Readiness, and Simulation-to-Real Gaps

Supporting material for *What the Survey Literature Says About Reinforcement Learning for Building Control: A Critical Synthesis*

Scope and coding scale

This supplementary file provides the survey-level evidence audit for RQ5. It reports validation modalities, validation rigor, deployment readiness, simulation-to-real treatment, gap signals, and concise survey-level answers.

Score	Evidence-intensity interpretation
0	Not addressed
1	Mentioned
2	Discussed
3	Structuring role

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S5.1 A review of reinforcement learning for autonomous building energy management [1]

Table S5.1: RQ5 recoded audit table for Mason and Grijalva (2019). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Mason and Grijalva (2019)	2	2	2	2	The review discusses validation maturity mainly through the dominance of simulated building energy-management problems, the dependence of RL evaluation on accurate simulator design, and the need for real-world-representative data. Deployment-readiness treatment receives score 2 because the paper explains that direct online learning in physical buildings would involve costly trial-and-error exploration, making simulation pretraining and later physical refinement important. Simulation-to-real discussion receives score 2 because the paper explicitly connects simulator fidelity, physical-building implementation, pretraining, and transfer learning. Gap-signal visibility receives score 2 because the paper clearly identifies simulation dependence and transfer to broader deployment as limitations, although it does not provide a full validation taxonomy or staged deployment ladder.

Table S5.2: RQ5 validation-modality scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Mason and Grijalva (2019)	3	0	0	1	1

Table S5.3: RQ5 validation-rigor scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Mason and Grijalva (2019)	1	2	2	1	1

Table S5.4: RQ5 gap-signal scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Mason and Grijalva (2019)	2	1	2	1	0

Table S5.5: RQ5 summary table for Mason and Grijalva (2019), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Mason and Grijalva (2019)	Strongly simulation-dominant with early deployment caution	Validation is framed mainly as simulated training followed, where possible, by later physical implementation or refinement. The key concern is that direct real-building learning is too costly during early exploration, so credible simulation and transfer become important.	The survey recognizes the simulation-to-real problem clearly, but it does not yet offer a well-developed validation ladder across testbeds, pilots, field trials, and long-term deployment maturity stages.

Table S5.6: One-line RQ5 answer for Mason and Grijalva (2019), after recoding.

Survey	RQ5 answer
Mason and Grijalva (2019)	Mason and Grijalva (2019) treat validation maturity mainly as a shift from simulation-trained RL toward limited physical implementation, emphasizing the need for accurate simulators, warning against costly direct exploration in real buildings, and pointing to transfer learning as a future route to wider deployment.

S5.2 Reinforcement learning for building controls: The opportunities and challenges [2]

Table S5.7: RQ5 recoded audit table for Wang and Hong (2020). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Wang and Hong (2020)	3	3	3	3	The review structurally distinguishes simulation environments from real buildings and reports that real-building deployment remains rare. Validation-maturity treatment receives score 3 because simulation-heavy evidence and limited real-building implementation are central to the paper’s assessment of RL for building controls. Deployment-readiness treatment receives score 3 because the paper explicitly organizes deployment barriers around time- and data-intensive training, control security and robustness, and limited generalization. Simulation-to-real discussion receives score 3 because the paper links simulated evaluation, real-building deployment limits, transfer learning, and the need for shared test infrastructure. Gap-signal visibility receives score 3 because open-source testbeds, datasets, benchmarking, and transfer learning are presented as structural needs for improving comparability and deployment readiness.

Table S5.8: RQ5 validation-modality scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Wang and Hong (2020)	3	2	0	1	2

Table S5.9: RQ5 validation-rigor scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Wang and Hong (2020)	2	3	3	2	1

Table S5.10: RQ5 gap-signal scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Wang and Hong (2020)	2	2	2	1	1

Table S5.11: RQ5 summary table for Wang and Hong (2020), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Wang and Hong (2020)	Simulation-dominant with explicit real-deployment barriers	Validation is framed through a strong simulation-versus-real-building contrast, with actual deployment treated as limited and blocked by training burden, security and robustness concerns, and weak generalization.	The survey gives a clear deployment-readiness critique rather than only a list of validation modes. Its main RQ5 value is in diagnosing why simulated RL success has not translated easily into real-building adoption and why open testbeds, datasets, and transfer learning are needed.

Table S5.12: One-line RQ5 answer for Wang and Hong (2020), after recoding.

Survey	RQ5 answer
Wang and Hong (2020)	Wang and Hong (2020) define validation maturity mainly through the contrast between simulation-heavy RL research and rare real-building deployment, arguing that deployment readiness is constrained by expensive training, insufficient robustness and security, and weak generalization, with transfer learning and shared test infrastructure proposed as partial remedies.

S5.3 The reinforcement learning method for occupant behavior in building control: A review [3]

Table S5.13: RQ5 recoded audit table for Han et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Han et al. (2021)	3	2	3	3	The review treats validation maturity as a central difficulty for occupant-behavior-oriented RL in building control. Validation-maturity treatment receives score 3 because the paper explicitly states that there is no general scientific standard for validating models when multiple occupant behaviors are involved. Deployment-readiness treatment receives score 2 because the paper discusses implementation difficulty, training requirements, robustness, and real implementations as a way to validate and improve methods, but deployment is not the main organizing axis. Simulation-to-real discussion receives score 3 because context-dependent occupant behavior and transfer from one setting to a more complex scenario are central validation concerns. Gap-signal visibility receives score 3 because lack of validation standards, transfer difficulty, context dependence, and generalization limits are explicit limitations in the paper.

Table S5.14: RQ5 validation-modality scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Han et al. (2021)	3	1	0	1	2

Table S5.15: RQ5 validation-rigor scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Han et al. (2021)	3	2	3	1	1

Table S5.16: RQ5 gap-signal scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Han et al. (2021)	1	1	3	1	1

Table S5.17: RQ5 summary table for Han et al. (2021), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Han et al. (2021)	Simulation-oriented with explicit concern about transfer and validation ambiguity	Validation is framed mainly through the absence of clear validation standards, the context dependence of occupant behavior, and the difficulty of transferring knowledge from one setting to another more complex one.	The survey contributes to RQ5 less by building a staged validation ladder and more by showing that occupant-centered RL faces a poorly standardized and transfer-sensitive path from simulated studies toward credible real-world use.

Table S5.18: One-line RQ5 answer for Han et al. (2021), after recoding.

Survey	RQ5 answer
Han et al. (2021)	Han et al. (2021) treat validation maturity mainly as a problem of weak validation standards, context-dependent occupant behavior, and limited transfer from simulated or simplified settings to more complex real-world environments, while only lightly touching actual implementation as a way to validate and improve RL methods.

S5.4 A review of deep reinforcement learning for smart building energy management [4]

Table S5.19: RQ5 recoded audit table for Yu et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Yu et al. (2021)	3	3	3	3	The review evaluates DRL methods in terms of convergence, effectiveness, scalability, and robustness, while also distinguishing model-free and model-based DRL in terms of interaction burden, exploration cost, and deployment feasibility. Validation-maturity treatment receives score 3 because implementation mode is structurally visible through the model-free/model-based comparison and tabulated practical-implementation evidence. Deployment-readiness treatment receives score 3 because the paper explicitly argues that fully exploring the state space in real buildings is impractical due to high cost and long training time, and it presents deployment-oriented pipelines involving building energy modeling, calibration, offline training, and real deployment. Simulation-to-real discussion receives score 3 because practical readiness is framed through calibrated models, offline training, domain randomization, transfer, and movement from virtual environments into real buildings. Gap-signal visibility receives score 3 because costly real-building exploration, simulation-heavy evaluation, and the need for model-based routes to deployment are central limitations and future-facing concerns in the survey.

Table S5.20: RQ5 validation-modality scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Yu et al. (2021)	3	1	1	2	2

Table S5.21: RQ5 validation-rigor scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Yu et al. (2021)	3	3	3	1	2

Table S5.22: RQ5 gap-signal scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Yu et al. (2021)	2	1	2	1	1

Table S5.23: RQ5 summary table for Yu et al. (2021), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Yu et al. (2021)	Simulation-dominant but deployment-structured	Validation is framed mainly through the contrast between model-free DRL, which is mostly simulation-evaluated, and model-based DRL, which is presented as the main route toward practical deployment through modeling, calibration, offline training, and transfer to real environments.	The survey answers RQ5 less by offering a full validation ladder and more by identifying a practical bridge from simulation to deployment, where model-based DRL serves as the main pathway for reducing costly and risky real-building exploration.

Table S5.24: One-line RQ5 answer for Yu et al. (2021), after recoding.

Survey	RQ5 answer
Yu et al. (2021)	Yu et al. (2021) treat validation maturity mainly through the difference between simulation-heavy model-free DRL and more deployment-oriented model-based DRL, arguing that real-building exploration is too costly and slow for direct training and that practical readiness therefore depends on calibrated models, offline training, and transfer into real environments.

S5.5 A review of reinforcement learning applications to control of heating, ventilation and air conditioning systems [5]

Table S5.25: RQ5 recoded audit table for Sierla et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Sierla et al. (2022)	3	3	2	3	The review treats validation maturity not only as a question of where an RL controller is tested, but also as a question of whether the control formulation is physically meaningful and deployable in HVAC systems. Validation-maturity treatment receives score 3 because the paper structurally analyzes action formulations, HVAC abstraction levels, and whether RL actions correspond to actuator commands, control-loop setpoints, or supervisory variables. Deployment-readiness treatment receives score 3 because the feasibility of interfacing RL outputs with actual HVAC equipment is central to the paper’s critique. Simulation-to-real discussion receives score 2 because the paper distinguishes physical HVAC systems from virtual environments such as building simulators and data-driven models, but it does not provide a full sim-to-real pipeline. Gap-signal visibility receives score 3 because vague action mappings, excessive abstraction, and unclear deployment interfaces are structural barriers identified by the review.

Table S5.26: RQ5 validation-modality scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Sierla et al. (2022)	2	1	0	1	0

Table S5.27: RQ5 validation-rigor scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Sierla et al. (2022)	3	3	2	1	1

Table S5.28: RQ5 gap-signal scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Sierla et al. (2022)	1	1	2	3	1

Table S5.29: RQ5 summary table for Sierla et al. (2022), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Sierla et al. (2022)	Deployment-conscious and interface-oriented, but still largely virtual-environment aware	Validation is framed mainly through whether the RL action space and abstraction level are realistic enough to interface with actual HVAC systems, rather than through a formal ladder from simulation to field deployment.	The survey contributes to RQ5 by showing that validation maturity depends not only on where the controller is tested, but also on whether the control formulation itself is physically meaningful and deployable in a BACS/HVAC context.

Table S5.30: One-line RQ5 answer for Sierla et al. (2022), after recoding.

Survey	RQ5 answer
Sierla et al. (2022)	Sierla et al. (2022) treat validation maturity mainly as a question of deployable control formulation, arguing that RL studies should be judged by whether their action spaces and abstraction levels can interface credibly with real HVAC systems, not just by simulated optimization performance.

S5.6 Applications of reinforcement learning for building energy efficiency control: A review [6]

Table S5.31: RQ5 recoded audit table for Fu et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Fu et al. (2022)	2	3	3	3	The review discusses validation maturity mainly through the practical difficulty of applying RL in real buildings and the continuing reliance on simulation platforms. Validation-maturity treatment receives score 2 because the paper discusses simulation platforms and some practical experiments or real-world implementations, but does not build a full validation taxonomy. Deployment-readiness treatment receives score 3 because the paper explicitly identifies real-building training as time-consuming, costly, inconvenient for occupants, potentially damaging to equipment, associated with uncontrollable costs, and affected by security concerns. Simulation-to-real discussion receives score 3 because the difference between simulated and real environments is named as a major obstacle. Gap-signal visibility receives score 3 because deployment cost, safety/security risk, training burden, environment modeling, and simulation-real mismatch are explicit barriers in the review’s challenge discussion.

Table S5.32: RQ5 validation-modality scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Fu et al. (2022)	3	1	1	1	0

Table S5.33: RQ5 validation-rigor scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Fu et al. (2022)	1	3	3	3	1

Table S5.34: RQ5 gap-signal scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Fu et al. (2022)	2	1	3	2	1

Table S5.35: RQ5 summary table for Fu et al. (2022), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Fu et al. (2022)	Simulation-dominant with explicit real-deployment obstacles	Validation is framed mainly through the practical difficulty of moving RL from simulation into real buildings, especially because trial-and-error training in real environments is slow, costly, risky for occupants and equipment, and affected by security concerns.	The survey contributes to RQ5 mainly by showing that deployment readiness is constrained less by lack of algorithmic promise than by the operational burden of real-world training and by unresolved differences between simulated and real environments.

Table S5.36: One-line RQ5 answer for Fu et al. (2022), after recoding.

Survey	RQ5 answer
Fu et al. (2022)	Fu et al. (2022) treat validation maturity mainly as a practical deployment problem, arguing that RL remains heavily simulation-based because direct training in real buildings is time-consuming, costly, potentially unsafe for occupants and equipment, and affected by a persistent gap between simulated and real environments.

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S5.7 Systematic review on deep reinforcement learning-based energy management for different building types [7]

Table S5.37: RQ5 recoded audit table for Shaqour and Hagishima (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Shaqour and Hagishima (2022)	3	3	3	3	The review treats validation maturity as a structurally visible issue in DRL-based BEMS research. Validation-maturity treatment receives score 3 because the paper reports the scarcity of real implementations and validations while also organizing reviewed studies by building type and validation setting. Deployment-readiness treatment receives score 3 because the paper emphasizes that pre-training or offline training before real-world deployment is critical, since untrained DRL agents are not suitable for direct deployment. Simulation-to-real discussion receives score 3 because the paper connects data hunger, offline training, scarce real-world validation, and the need to move beyond purely simulated evaluation. Gap-signal visibility receives score 3 because limited real-system validation, data requirements, and deployment-readiness barriers are explicit recurring limitations, even though laboratory, university-building, school, and other real-system examples are beginning to appear.

Table S5.38: RQ5 validation-modality scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Shaqour and Hagishima (2022)	3	2	2	1	1

Table S5.39: RQ5 validation-rigor scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Shaqour and Hagishima (2022)	3	3	3	1	2

Table S5.40: RQ5 gap-signal scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.					
Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Shaqour and Hagishima (2022)	2	2	2	1	2

Table S5.41: RQ5 summary table for Shaqour and Hagishima (2022), after recoding.			
Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Shaqour and Hagishima (2022)	Simulation-dominant with explicit concern about real validation scarcity	Validation is framed mainly through the need for sufficient offline training before deployment and through the observation that real implementations remain rare, although laboratory, school, university-building, and other real-system validations provide early non-simulation evidence.	The survey contributes to RQ5 by showing that deployment readiness in DRL-based BEMS is strongly constrained by data availability and pre-training requirements, while also indicating that intermediate real-system validation forms are beginning to appear.

Table S5.42: One-line RQ5 answer for Shaqour and Hagishima (2022), after recoding.	
Survey	RQ5 answer
Shaqour and Hagishima (2022)	Shaqour and Hagishima (2022) treat validation maturity mainly as a problem of moving data-hungry DRL methods beyond simulation through sufficient offline training and scarce real-system validation, while noting that laboratory, school, university-building, and other real-system experiments provide early but still limited evidence of deployment readiness.

S5.8 Building energy management with reinforcement learning and model predictive control: A survey [8]

Table S5.43: RQ5 recoded audit table for Zhang et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Zhang et al. (2022)	2	3	3	3	The review discusses validation maturity mainly through real-world feasibility rather than through a formal validation taxonomy. Validation-maturity treatment receives score 2 because the paper discusses the limitations of direct RL exploration in buildings, but does not structure the review around validation stages. Deployment-readiness treatment receives score 3 because real-world applicability is explicitly framed around data efficiency, safety, robust adaptability, high economic cost from bad exploratory actions, and long real-world training time. Simulation-to-real discussion receives score 3 because the paper presents model-based RL, learning from historical data, and MPC/RL combinations as practical routes for reducing unsafe or inefficient real-building exploration. Gap-signal visibility receives score 3 because safety, data efficiency, robust adaptability, and real-world feasibility are explicit deployment barriers and future-facing limitations in the review.

Table S5.44: RQ5 validation-modality scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Zhang et al. (2022)	2	1	0	1	2

Table S5.45: RQ5 validation-rigor scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Zhang et al. (2022)	1	3	3	3	1

Table S5.46: RQ5 gap-signal scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Zhang et al. (2022)	1	1	2	2	0

Table S5.47: RQ5 summary table for Zhang et al. (2022), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Zhang et al. (2022)	Deployment-conscious and hybrid-method oriented	Validation is framed mainly through real-world feasibility, with direct RL deployment treated as costly and risky, and model-based or MPC-assisted approaches presented as more practical routes toward usable building control.	The survey contributes to RQ5 by recasting the simulation-to-real gap as a practical control-design problem centered on data efficiency, safety, and robust adaptability rather than as a formal maturity ladder across validation stages.

Table S5.48: One-line RQ5 answer for Zhang et al. (2022), after recoding.

Survey	RQ5 answer
Zhang et al. (2022)	Zhang et al. (2022) treat validation maturity mainly as a question of real-world feasibility, arguing that direct RL in buildings is limited by expensive exploration, safety concerns, and weak robust adaptability, so stronger deployment readiness depends on model-based RL, historical data use, and MPC-assisted or hybrid control designs.

S5.9 Reinforcement learning: theory and applications in HEMS [9]

Table S5.49: RQ5 recoded audit table for Al-Ani and Das (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Al-Ani and Das (2022)	1	1	1	1	The paper is a broad review of reinforcement learning theory and applications in home energy management systems. Validation-maturity treatment receives score 1 because the paper mentions that RL research in HEMS is still at an early stage and that actual HEMS deployment remains limited, but validation maturity does not structure the review. Deployment-readiness treatment receives score 1 because the paper distinguishes offline model-based training from online interaction with the real physical environment, but does not develop a detailed deployment-readiness framework. Simulation-to-real discussion receives score 1 because the paper notes the use of HEMS models and simulation-based training, but does not provide a sustained simulation-to-real analysis. Gap-signal visibility receives score 1 because limited deployment and early-stage maturity are acknowledged, while the paper’s main structure remains focused on RL methods, HEMS application classes, and performance metrics rather than validation or deployment gaps.

Table S5.50: RQ5 validation-modality scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Al-Ani and Das (2022)	2	0	0	1	0

Table S5.51: RQ5 validation-rigor scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Al-Ani and Das (2022)	1	1	1	0	1

Table S5.52: RQ5 gap-signal scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.					
Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Al-Ani and Das (2022)	1	1	0	0	0

Table S5.53: RQ5 summary table for Al-Ani and Das (2022), after recoding.			
Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Al-Ani and Das (2022)	Broad and lightly deployment-aware, but still model/simulation-oriented	Validation is framed mainly at a high level through the distinction between offline model-based training and online interaction with the real environment, together with the observation that actual HEMS deployment remains limited in prior literature.	The survey contributes to RQ5 mainly by confirming that RL for HEMS remains early-stage and still largely trained through models, rather than by developing a strong validation-maturity ladder or a systematic simulation-to-real critique.

Table S5.54: One-line RQ5 answer for Al-Ani and Das (2022), after recoding.	
Survey	RQ5 answer
Al-Ani and Das (2022)	Al-Ani and Das (2022) treat validation maturity mainly as an early-stage HEMS literature issue, acknowledging limited reported deployment and widespread model-based training, but focusing more on broad method coverage, HEMS application classes, and performance metrics than on a detailed deployment-readiness or simulation-to-real framework.

S5.10 A review of reinforcement learning for controlling building energy systems from a computer science perspective [10]

Table S5.55: RQ5 recoded audit table for Weinberg et al. (2023). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Weinberg et al. (2023)	2	3	3	3	The review is a computer-science-oriented survey of reinforcement learning for building energy systems. Validation-maturity treatment receives score 2 because the paper does not build a staged validation taxonomy, but discusses practical readiness through sample efficiency, environment-interaction burden, transfer, scalability, and risky exploration. Deployment-readiness treatment receives score 3 because deployment readiness is structurally tied to the paper’s computer-science challenge framework, including sample efficiency, transfer between buildings, scalability, multi-agent control, communication, and system design. Simulation-to-real discussion receives score 3 because transferability, sample efficiency, environment-interaction burden, and model-based reinforcement learning are central to how the paper frames practical use. Gap-signal visibility receives score 3 because the paper is explicitly organized around research gaps and future directions in these computer-science challenges.

Table S5.56: RQ5 validation-modality scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Weinberg et al. (2023)	1	0	0	0	2

Table S5.57: RQ5 validation-rigor scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Weinberg et al. (2023)	1	3	3	2	0

Table S5.58: RQ5 gap-signal scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Weinberg et al. (2023)	1	0	3	2	0

Table S5.59: RQ5 summary table for Weinberg et al. (2023), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Weinberg et al. (2023)	Technical-readiness oriented, with strong emphasis on transfer and sample efficiency	Validation is framed mainly through the technical conditions needed for practical reinforcement learning in building energy systems, especially sample efficiency, reduced environment interaction, transfer of policies between buildings, scalability, and multi-agent coordination, rather than through an explicit ladder from simulation to field deployment.	The survey contributes to RQ5 by showing that deployment maturity depends heavily on whether reinforcement learning can become data-efficient, transferable, scalable, and safe enough for real building energy system operation, even though the paper is less focused on direct field-validation categories.

Table S5.60: One-line RQ5 answer for Weinberg et al. (2023), after recoding.

Survey	RQ5 answer
Weinberg et al. (2023)	Weinberg et al. (2023) treat validation maturity mainly as a problem of technical readiness for practical building energy system control, arguing that wider real-world deployment requires stronger sample efficiency, reduced risky environment interaction, transferability across buildings, scalability, and multi-agent coordination, rather than merely more simulation studies.

S5.11 Dynamic indoor thermal environment using reinforcement learning-based controls: Opportunities and challenges [11]

Table S5.61: RQ5 recoded audit table for Chatterjee and Khovalyg (2023). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Chatterjee and Khovalyg (2023)	3	3	3	3	The review treats validation and deployment maturity as structurally visible issues in reinforcement-learning-based HVAC control. Validation-maturity treatment receives score 3 because the paper summarizes simulation software, building model scale, building type, and real-building implementation status. Deployment-readiness treatment receives score 3 because the conclusion explicitly states that reinforcement-learning controllers remain largely in the research-and-development stage, with limited acceptance in actual buildings. Simulation-to-real discussion receives score 3 because the paper contrasts simulation-trained controllers with limited real-building implementation and identifies training burden, control security, and transfer learning as major deployment obstacles. Gap-signal visibility receives score 3 because the paper explicitly calls for data-rich, open-source, interoperable virtual testbeds for cross-study validation and benchmarking.

Table S5.62: RQ5 validation-modality scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Chatterjee and Khovalyg (2023)	3	1	1	2	1

Table S5.63: RQ5 validation-rigor scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Chatterjee and Khovalyg (2023)	3	3	3	2	2

Table S5.64: RQ5 gap-signal scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Chatterjee and Khovalyg (2023)	2	2	3	2	2

Table S5.65: RQ5 summary table for Chatterjee and Khovalyg (2023), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Chatterjee and Khovalyg (2023)	Simulation-dominant but explicitly deployment-aware	Validation is framed through a strong contrast between simulation-trained reinforcement-learning controllers and their still-limited use in actual buildings, with deployment readiness tied to training burden, control security, transfer learning, and better shared virtual test infrastructure.	The survey contributes to RQ5 by making the reality-gap problem unusually explicit: most reinforcement-learning studies are still simulation-trained, only a minority reach real buildings, and stronger deployment maturity will require safer, more data-efficient, and more transferable control pipelines.

Table S5.66: One-line RQ5 answer for Chatterjee and Khovalyg (2023), after recoding.

Survey	RQ5 answer
Chatterjee and Khovalyg (2023)	Chatterjee and Khovalyg (2023) treat validation maturity mainly as the problem of moving reinforcement-learning controllers beyond simulation into real buildings, arguing that the field remains largely simulation-trained and that wider deployment is still constrained by training cost, control security, limited transferability, and the lack of shared validation infrastructure.

S5.12 Reinforcement learning for HVAC control in intelligent buildings: A technical and conceptual review [12]

Table S5.67: RQ5 recoded audit table for Al Sayed et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Al Sayed et al. (2024)	3	3	3	3	The review treats virtual-to-real deployment as a structuring concern for reinforcement-learning-based HVAC control. Validation-maturity treatment receives score 3 because the paper reports limited real-building deployment and structurally reviews simulation environments, model environments, and training episode design. Deployment-readiness treatment receives score 3 because the paper identifies unsafe direct real use, insufficient exogenous-state diversification, and costly retraining or fine-tuning as central barriers to practical deployment. Simulation-to-real discussion receives score 3 because the paper directly addresses virtual training environments, real-world distribution shift, changing occupancy schedules, electricity prices, and weather, and the risk that learned virtual distributions become outdated before deployment. Gap-signal visibility receives score 3 because weak virtual-to-real transfer, limited training diversity, retraining cost, and the need for meta-reinforcement-learning adaptation are structural limitations and future directions in the review.

Table S5.68: RQ5 validation-modality scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Al Sayed et al. (2024)	3	0	1	1	2

Table S5.69: RQ5 validation-rigor scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Al Sayed et al. (2024)	3	3	3	3	1

Table S5.70: RQ5 gap-signal scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.					
Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Al Sayed et al. (2024)	2	2	3	3	2

Table S5.71: RQ5 summary table for Al Sayed et al. (2024), after recoding.			
Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Al Sayed et al. (2024)	Simulation-dominant but strongly deployment-aware	Validation is framed mainly through the gap between virtual training environments and non-stationary real-world deployment, with readiness depending on whether training episodes capture sufficient diversity in exogenous state components and whether retraining can be reduced.	The survey contributes to RQ5 by showing that deployment maturity is blocked less by lack of algorithmic promise than by unsafe direct transfer from simulation, weak training diversity, distribution shift, and expensive retraining under real-world change.

Table S5.72: One-line RQ5 answer for Al Sayed et al. (2024), after recoding.	
Survey	RQ5 answer
Al Sayed et al. (2024)	Al Sayed et al. (2024) treat validation maturity mainly as a problem of safely moving simulation-trained HVAC controllers into non-stationary real buildings, arguing that limited real deployment, insufficiently diverse training episodes, distribution shift, and costly retraining make generalization the main barrier, with meta-reinforcement learning proposed as a promising remedy.

S5.13 An overview of reinforcement learning-based approaches for smart home energy management systems with energy storages [13]

Table S5.73: RQ5 recoded audit table for Pinthurat et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Pinthurat et al. (2024)	3	3	3	3	The review treats real-world testing and validation as a structuring future-research need for reinforcement-learning-based smart home energy management systems with energy storage. Validation-maturity treatment receives score 3 because the paper includes a dedicated discussion of real-world testing and validation as necessary for assessing performance, adaptability, scalability, and practical viability. Deployment-readiness treatment receives score 3 because the paper explicitly connects deployment readiness to field trials under real user behavior, weather, and grid-dynamic conditions. Simulation-to-real discussion receives score 3 because validation in real environments is presented as necessary to refine algorithms, validate models, optimize parameters, and bridge theory and practice. Gap-signal visibility receives score 3 because lack of real-world testing, field validation, and practical deployment evidence is an explicit future-research direction, alongside concerns about data intensity, cost, safety, privacy, and security.

Table S5.74: RQ5 validation-modality scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Pinthurat et al. (2024)	2	0	1	2	2

Table S5.75: RQ5 validation-rigor scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Pinthurat et al. (2024)	1	3	3	2	1

Table S5.76: RQ5 gap-signal scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Pinthurat et al. (2024)	1	3	3	2	2

Table S5.77: RQ5 summary table for Pinthurat et al. (2024), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Pinthurat et al. (2024)	Future-oriented and deployment-aware, but still field-evidence limited	Validation is framed mainly as a missing practical step: the paper emphasizes the need for real-world testing and field validation to examine adaptability, scalability, user behavior, weather variation, grid dynamics, and practical viability.	The survey contributes to RQ5 by arguing that reinforcement-learning-based smart-home energy management with storage still needs much stronger real-world validation before deployment readiness can be claimed with confidence. Its strongest contribution is not a validation taxonomy, but a clear future agenda for field testing and theory-to-practice validation.

Table S5.78: One-line RQ5 answer for Pinthurat et al. (2024), after recoding.

Survey	RQ5 answer
Pinthurat et al. (2024)	Pinthurat et al. (2024) treat validation maturity mainly as a gap between promising RL-based smart-home energy management methods and the still-limited real-world testing needed to verify performance, adaptability, scalability, and practical viability under real user, weather, and grid conditions.

S5.14 Applications of Deep Reinforcement Learning for Home Energy Management Systems: A Review [14]

Table S5.79: RQ5 recoded audit table for Latoń et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Latoń et al. (2024)	3	3	3	3	The review treats validation maturity and practical implementation as structurally visible issues for reinforcement learning and deep reinforcement learning in home energy management systems. Validation-maturity treatment receives score 3 because the paper explicitly contrasts theoretical or simulation-based research with the need for practical case studies and real-world implementations, and its tabulated review shows simulation-heavy verification, limited real implementation, and use of real-world data in only some cases. Deployment-readiness treatment receives score 3 because the paper discusses practical implementation barriers including data intensity, safety and reliability, scalability, heterogeneous infrastructure, privacy, and security. Simulation-to-real discussion receives score 3 because the paper frames maturity as a theory-to-practice gap that requires testing under real conditions and translating simulation results into operational evidence. Gap-signal visibility receives score 3 because lack of real applications, lack of detailed case studies, limited tangible operational impact, security/privacy issues, and scalability constraints are explicit gaps and future needs.

Table S5.80: RQ5 validation-modality scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Latoń et al. (2024)	3	0	1	1	2

Table S5.81: RQ5 validation-rigor scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Latoń et al. (2024)	3	3	3	3	1

Table S5.82: RQ5 gap-signal scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Latoń et al. (2024)	3	3	2	2	3

Table S5.83: RQ5 summary table for Latoń et al. (2024), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Latoń et al. (2024)	Simulation-dominant but explicitly practice-oriented	Validation is framed mainly through the gap between simulation-heavy RL/DRL research and the need for practical case studies and real-world implementations, with deployment readiness linked to data pipelines, safety, reliability, scalability, privacy, security, and heterogeneous system integration.	The survey contributes to RQ5 by showing that maturity is limited less by lack of promising algorithms than by the shortage of real-world case evidence and by the broader practical conditions required for secure, reliable, scalable, and operational deployment.

Table S5.84: One-line RQ5 answer for Latoń et al. (2024), after recoding.

Survey	RQ5 answer
Latoń et al. (2024)	Latoń et al. (2024) treat validation maturity mainly as a theory-to-practice gap, arguing that RL/DRL for HEMS remains largely simulation-based and that stronger real-world case studies, together with scalable, safe, reliable, and privacy-preserving deployment infrastructures, are needed before deployment readiness can be claimed confidently.

S5.15 Multi objectives reinforcement learning for smart buildings: A systematic review of algorithms, applications and future perspectives [15]

Table S5.85: RQ5 recoded audit table for Huynh et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Huynh et al. (2025)	2	3	2	3	The review is a systematic review of multi-objective reinforcement learning in smart buildings. Validation-maturity treatment receives score 2 because the paper discusses open-source infrastructures, simulation-oriented environments, and more realistic testbeds, but it does not organize the review as a full validation-maturity taxonomy. Deployment-readiness treatment receives score 3 because the paper repeatedly identifies practical deployment barriers, including computational complexity, long training times, high-quality data needs, heterogeneity across building types and HVAC configurations, interpretability, transparency, and stakeholder trust. Simulation-to-real discussion receives score 2 because the paper discusses simulation/testbed infrastructure and generalization across heterogeneous smart-building settings, but direct field deployment evidence remains weak. Gap-signal visibility receives score 3 because scalability, data requirements, heterogeneity, computational feasibility, interpretability, transparency, and trust are explicit barriers and future directions for real-world MORL adoption.

Table S5.86: RQ5 validation-modality scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Huynh et al. (2025)	3	2	0	0	2

Table S5.87: RQ5 validation-rigor scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Huynh et al. (2025)	2	3	2	2	0

Table S5.88: RQ5 gap-signal scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Huynh et al. (2025)	2	1	2	1	1

Table S5.89: RQ5 summary table for Huynh et al. (2025), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Huynh et al. (2025)	Deployment- and generalization-aware, but field-validation thin	Validation is framed mainly through the practical conditions needed for real-world MORL deployment, especially computational feasibility, data sufficiency, robustness, heterogeneity across building types and HVAC configurations, interpretability, transparency, stakeholder trust, and simulation/testbed infrastructure.	The survey contributes to RQ5 by showing that maturity in MORL for smart buildings depends heavily on whether methods can scale, generalize across diverse buildings, remain understandable and computationally feasible, and move beyond simulation-oriented infrastructure toward stronger real-world validation.

Table S5.90: One-line RQ5 answer for Huynh et al. (2025), after recoding.

Survey	RQ5 answer
Huynh et al. (2025)	Huynh et al. (2025) treat validation maturity mainly as a practical deployment and generalization problem, arguing that MORL for smart buildings still faces barriers in computational complexity, training burden, data needs, cross-building heterogeneity, interpretability, transparency, and trust before it can be considered ready for broader real-world use.

S5.16 Reinforcement learning for control and optimization of real buildings: Identifying and addressing implementation hurdles [16]

Table S5.91: RQ5 recoded audit table for Kannari et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Kannari et al. (2025)	3	3	3	3	The review is explicitly focused on reinforcement-learning-based control and optimization in real buildings. Validation-maturity treatment receives score 3 because the paper summarizes practical deployments, implementation settings, deployment modes, and how thoroughly existing applications have been tested. Deployment-readiness treatment receives score 3 because the paper discusses pretraining, static versus dynamic deployment, online relearning, recalibration, safety checks, fallback to traditional control, and validation under varying real-world conditions. Simulation-to-real discussion receives score 3 because the paper explains that real-world reinforcement-learning controllers commonly rely on virtual pretraining before deployment, and it discusses digital twins as data-rich platforms for pre-deployment training and operational adaptation. Gap-signal visibility receives score 3 because the paper explicitly identifies limitations around long-term real-world evidence, scalable deployment, real-world comparability under varying weather and occupancy, and the need for robust validation procedures.

Table S5.92: RQ5 validation-modality scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Kannari et al. (2025)	2	3	2	3	2

Table S5.93: RQ5 validation-rigor scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Kannari et al. (2025)	3	3	3	3	2

Table S5.94: RQ5 gap-signal scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Kannari et al. (2025)	1	0	2	2	3

Table S5.95: RQ5 summary table for Kannari et al. (2025), after recoding.

Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Kannari et al. (2025)	Strongly deployment- and validation-procedure oriented	Validation is framed through actual real-building implementation practice, including virtual pretraining, static versus dynamic deployment, online relearning, recalibration, safety checks, fallback control, digital twins, and validation under varying real-world conditions.	The survey contributes to RQ5 by moving beyond general deployment concerns and showing concretely how reinforcement-learning controllers are pretrained, safeguarded, deployed, and validated in real buildings, while also showing that short and heterogeneous test durations leave major uncertainty about long-term reliability, scalability, and comparability.

Table S5.96: One-line RQ5 answer for Kannari et al. (2025), after recoding.

Survey	RQ5 answer
Kannari et al. (2025)	Kannari et al. (2025) treat validation maturity mainly as a real-building implementation problem, arguing that reinforcement-learning controllers become more deployment-ready only when they are pretrained safely, equipped with safeguards and fallback logic, validated under varying real-world conditions, and supported by longer-term evidence and digital-twin infrastructure.

S5.17 Reinforcement learning for optimizing renewable energy utilization in buildings: A review on applications and innovations [17]

Table S5.97: RQ5 recoded audit table for Michailidis et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	Validation-maturity treatment	Deployment-readiness treatment	Simulation-to-real discussion	Gap-signal visibility	Recoded evidence justification
Michailidis et al. (2025)	2	2	2	1	The review analyzes reinforcement-learning applications in renewable-energy-integrated building energy management systems using categories that include training methodologies, testbed characteristics, application type, and implementation setting. Validation-maturity treatment receives score 2 because simulated and real-world implementations are made visible through the review design and summary tables, but validation maturity does not structure the paper as a dedicated framework. Deployment-readiness treatment receives score 2 because practical implementation setting is included and both simulated and real-life applications are acknowledged, but the paper does not develop a sustained deployment-readiness critique. Simulation-to-real discussion receives score 2 because the distinction between simulation and real-life application appears in the review categories, although without a strong simulation-to-real methodology discussion. Gap-signal visibility receives score 1 because validation-related gaps are only weakly signaled compared with the paper's stronger focus on RES integration, reward design, control methods, and application trends.

Table S5.98: RQ5 validation-modality scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Simulation	Emulation / testbed	Pilot / demonstration	Real building / field deployment	Cross-site / transfer / generalization evidence
Michailidis et al. (2025)	2	1	1	1	1

Table S5.99: RQ5 validation-rigor scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Validation taxonomy clarity	Deployment-readiness discussion	Simulation-to-real gap rigor	Safety / safeguards for deployment	Strength of real-world evidence
Michailidis et al. (2025)	2	2	2	0	1

Table S5.100: RQ5 gap-signal scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.					
Survey	Overreliance on simulation	Lack of pilots / field tests	Weak sim-to-real methodology	Limited deployment protocol detail	Limited long-term real-world evidence
Michailidis et al. (2025)	1	0	0	0	0

Table S5.101: RQ5 summary table for Michailidis et al. (2025), after recoding.			
Survey	Validation orientation	Dominant validation framing	Main RQ5 consequence
Michailidis et al. (2025)	Broadly implementation-aware, but validation-light	Validation is framed mainly through application-type, implementation-setting, and testbed descriptors in the review tables, with both simulated and real-life implementations acknowledged, rather than through a strong staged ladder from simulation to field maturity.	The survey contributes to RQ5 by making implementation setting visible across the reviewed renewable-energy-integrated BEMS literature, but it is weaker than the strongest RQ5 papers on explicit deployment-readiness critique, safety, long-term validation, and simulation-to-real diagnosis.

Table S5.102: One-line RQ5 answer for Michailidis et al. (2025), after recoding.	
Survey	RQ5 answer
Michailidis et al. (2025)	Michailidis et al. (2025) treat validation maturity mainly as an implementation-setting issue by distinguishing simulated and real-life reinforcement-learning applications and including testbed characteristics in their review design, but they contribute more to broad mapping of renewable-energy-integrated BEMS contexts than to a strong framework for deployment readiness or the simulation-to-real gap.

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